Who builds what, and in which order

Deep Learning Based Super-Resolution Mapping — 4× sub-pixel land cover maps at 2.5 m from 10 m Sentinel-2 imagery.

The scaffold already stands: model architecture, ingestion, API, dual-canvas UI, Docker and a passing constraint suite. These six roles are therefore split by what is **missing**, not by generic job titles — and two of them hold up everyone else.

ALREADY BUILT	STILL MISSING
• D-SUN unmixing, Swin allocation head, mass-conserving sub-pixel assignment, MRF smoothing • Tiled inference — a 256×256 tile in 4.8 s on CPU, mass error 2.4e-07 • STAC ingestion, SWIR band alignment, SCL cloud masking • Ingest → GPU tensor hand-off, COG / GeoJSON / CSV export • FastAPI routes, PostGIS schema, dual-canvas UI, Compose for GPU and CPU	• No trained weights and no training pipeline at all • No ground-truth labels — WorldCover → 5 classes → abundances is unwritten • Left canvas is fed a preview PNG where a tile template is expected • Job state lives in in-process dicts; srm_jobs is never written • mIoU is never computed; no baseline comparison against bicubic • data_cache/ is empty; Docker images have never been built

> **The one thing to get right.** Two of six people are on the critical path, and their work is the least visible. There is a real pull toward staffing four people on the UI because progress there is easy to see. Resist it — a beautiful interface rendering noise loses to a plain one rendering a correct map.

Sequencing

Hours from your build start. Labels gate training, training gates every visual result, and everything else proceeds without either.

H0–H2	Everyone clones, runs the CPU stack and sees the UI, so nobody is reasoning about code they have not watched run. QA starts the GPU image build immediately — longest pole, most likely to break.
H2–H10	Data and DL engineers pair on the label pipeline. Two people here is not redundancy, it is the whole schedule. Backend and frontend work their own fixes; neither needs weights.
H10–H24	Training runs, checkpointing hourly. The first checkpoint lets frontend and QA test against real output instead of noise. Backend finishes persistence and the mIoU path.
H24–H36	Integration. Real weights meet the real UI — expect bugs neither half showed alone. Cache the offline regions and verify with the network off.
H36–H44	Freeze. No new features. Time the three-minute script repeatedly, on the machine you will present from and the network you will actually have.
H44–H48	Buffer. If you are writing code here, something upstream slipped. Deck polish, fallback drills, sleep.

CRITICAL PATH

Deep Learning Engineer

OWNS

`ml_engine/models/` · `ml_engine/utils/unmixing_loss.py` · `ml_engine/train.py` · `ml_engine/weights/`

WORK

- Write **train.py** and the dataset loader. Neither exists yet — this is the single largest missing piece in the repository.
- Start from SEN2SRLite weights as the spatial backbone instead of training from scratch; only the allocation head genuinely needs to learn.
- Tune the four loss weights in **SRMLoss**. Watch that mass_error stays below 1e-3 — that constraint is the project's whole argument.
- Export a checkpoint every hour, however undertrained. Four people are blocked until a loadable weights file exists.

DONE WHEN

sih26142_srm_v1.pth exists, loads without shape errors, and produces a land cover map that visibly beats the 10 m input.

> **Watch out.** load_model uses strict=False, so mismatched state_dict keys fail **silently** and look exactly like a successful load of random weights. Verify parameters actually populated.

CRITICAL PATH

Geospatial Data Engineer

OWNS

`backend/services/stac_fetcher.py` · `backend/services/preprocessor.py` · `scripts/build_offline_cache.py` · `scripts/build_labels.py`

WORK

- Build the label pipeline **first**: ESA WorldCover's 11 classes remapped to our 5, then aggregated into abundance fractions. Training cannot start without it.
- Pair those with downsampled high-resolution reference imagery for the sub-pixel supervision signal.
- Run build_offline_cache.py for all four regions and verify it works with the network physically unplugged.
- Sanity-check band registration: no pixel offset between the native 10 m channels and the resampled 20 m SWIR.

DONE WHEN

The DL engineer has paired (X, abundance, fine_labels) tensors on disk, and data_cache/ holds four real regions.

> **Watch out.** Chennai is the only cached region outside UTM 43N. Cache it — a reprojection bug is invisible until a granule crosses zones, and it would surface live in front of judges.

PARALLEL

Backend & Spatial DB Lead

OWNS

`backend/routers/` · `backend/services/tasks.py` · `backend/database/` · `backend/services/exporter.py`

WORK

- Wire the job lifecycle to PostGIS. _JOBS, _TASK_IDS and _GRANULE_BBOX are in-process dictionaries that any restart wipes.
- Persist ClassMetrics and COGExport rows so the analytics survive too.
- Write the mIoU evaluation path, so the accuracy figure in the PRD is measured rather than asserted.
- Verify the QGIS round-trip yourself — the affine scaling on export is a classic silent failure.

DONE WHEN

A job survives an API restart, and a judge can open the exported GeoTIFF in QGIS in the right place on Earth.

> **Watch out.** A judge asking “how did you measure 0.70 mIoU?” is a bad moment if nothing computes it.

PARALLEL

Frontend Web GIS Developer

OWNS

`frontend/src/components/` · `frontend/src/store/` · `frontend/src/lib/`

WORK

- Fix the left canvas. It receives granule.preview_url, a rendered PNG, where an XYZ tile template is expected, so input imagery never draws. Point it at TiTiler.
- Add real loading and error states; a failed job currently just prints a red bar.
- Profile the curtain drag and the camera lock under fast panning; the sync guard must hold at 45 FPS.
- Keep the class palette in constants.js in step with the one in ml_engine/utils/cog_writer.py.

DONE WHEN

A judge draws a box, hits Execute, and watches a real 2.5 m map slide in under the curtain at 45 FPS.

> **Watch out.** Box drawing, region selection and the curtain drag were all broken and are now fixed — re-test them after any refactor of DualCanvasMap.

PARALLEL

Integration & QA Lead

OWNS

`docker-compose*.yml · ml_engine/tests/ · backend/tests/ · data_cache/`

WORK

- Build the Docker images on the RTX 3080 **today**. They have never been built once — assume something breaks and find out now, not on stage.
- Run scripts/check_gpu.py inside the container; record peak VRAM and tune MAX_PATCH_SIZE to fit.
- Own the end-to-end drill: cold clone, compose up, draw an AOI, export, open the result in QGIS.
- Rehearse the failure modes deliberately — pull the network, force an OOM, pick a heavily clouded scene.

DONE WHEN

A clean clone reaches a working demo on the 3080 with one command, and you have timed the full run three times.

> **Watch out.** The CPU override exists for machines without an NVIDIA runtime, but a stale DEVICE=cpu in .env will silently win on the GPU box too.

INTEGRATION OWNER

Team Lead & Presenter

OWNS

`docs/DEMO_SCRIPT.md · docs/PRD.md · README.md · the pitch deck`

WORK

- Guard the scope. Every P2 idea that appears at hour 30 is a threat, not an opportunity.
- Own the mass-conservation defence: when a judge asks whether this is generative invention, show that the output downsamples back to the observed sensor abundances.
- Never present a random-weight output as a result.
- Merge and integrate. With six people on one repository, nobody else should be resolving conflicts.

DONE WHEN

The three-minute run is rehearsed end to end, and you can answer the hallucination question without opening an editor.

> **Watch out.** The demo script in the source PRD said “Delhi Coastal/Port”. Delhi is landlocked; Chennai is the region that sentence wanted.

Hand-off contracts

Agree these in the first hour. Each one is a place where two people can lose half a day discovering they assumed different things.

FROM	TO	THE CONTRACT
Data eng.	**DL eng.**	X (6,H,W) float32 · abundance (5,H,W) summing to 1 · fine_labels (H·4,W·4) int64. Band order B02 B03 B04 B08 B11 B12 — fix it once, in writing.
DL eng.	**Backend**	Weights path and state_dict key names. strict=False means a mismatch loads silently as random weights.
Backend	**Frontend**	SRMResponse in backend/schemas.py is the contract. The UI polls /api/v1/jobs/{id} until status leaves PENDING. Do not rename fields without telling frontend.
Ingest	**GPU worker**	Documented in docs/PIPELINE_HANDOFF.md. The .npz keys must stay in step across two container images.
Everyone	**Team lead**	Branch per role, small commits, no direct pushes to main during integration hours.

```
Repository state verified 2026-08-26: 7 tests passing, both Compose configurations valid, frontend rendering with no console errors, 256×256 CPU inference at 4.8 s. Weights, labels and the training pipeline do not yet exist.
```